



كلية تكنولوجيا الصناعة والطاقة

Information Technology Department



جامعة برج العرب التكنولوجية

BORG AL ARAB TECHNOLOGICAL UNIVERSITY

INTRODUCTION TO C PROGRAMMING ESSENTIALS

First Year – Information Technology Program

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LECTURE 3

C LANGUAGE

BASICS

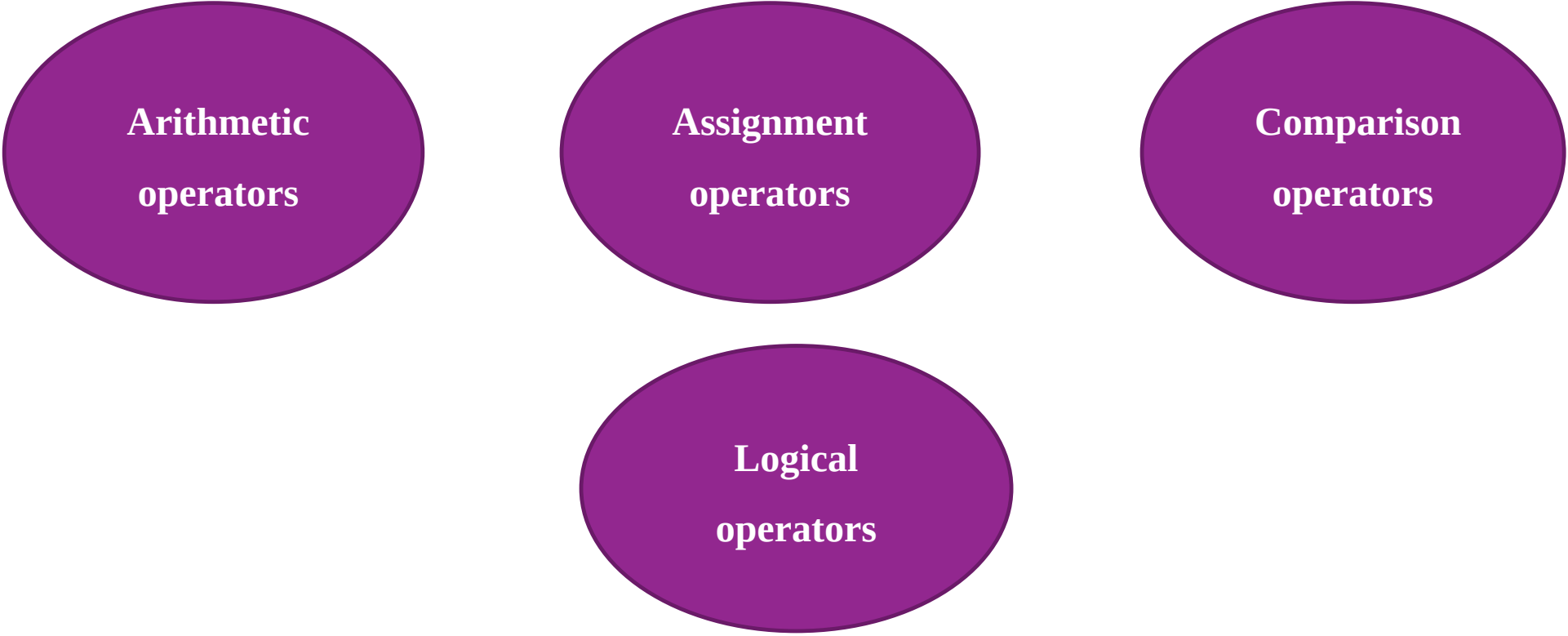
OPERATORS

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OPERATORS

Operators are used to perform operations on variables and values. C divides the operators into the following groups:



Arithmetic
operators

Assignment
operators

Comparison
operators

Logical
operators

ARITHMETIC OPERATORS

Arithmetic operators are used to perform common mathematical operations.

Operator	Name	Description	Example
+	Addition	Adds together two values	$x + y$
-	Subtraction	Subtracts one value from another	$x - y$
*	Multiplication	Multiplies two values	$x * y$
/	Division	Divides one value by another	x / y
%	Modulus	Returns the division remainder	$x \% y$
++	Increment	Increases the value of a variable by 1	<code>++x</code>
--	Decrement	Decreases the value of a variable by 1	<code>--x</code>

EXAMPLE

```
#include <stdio.h>

int main() {
    int x = 10;
    int y = 3;

    printf("%d\n", x + y);
    printf("%d\n", x - y);
    printf("%d\n", x * y);
    printf("%d\n", x / y);
    printf("%d\n", x % y);

    return 0;
}
```

Output

```
13
7
30
3
1
```

```
=== Code Execution Successful ===
```

NOTES

- When dividing two integers in C, the result will also be an integer. For example, $10 / 3$ gives 3. If you want a decimal result, use float or double values, like $10.0 / 3$.

Note:

- if you divide two integers: 5 by 2, you know that the result is 2.5. The result will only display the number 2 ?

```
float div = 5 / 2;
```

```
printf("%f", div ); // 2.000000
```

it is because 5 and 2 are still integers in the division. In this case, you need to manually convert the integer values to floating-point values.

```
// Manual conversion: int to float
```

```
float div = (float) 5 / 2;
```

```
printf("%f", div ); // 2.500000
```

INCREMENT

1) Postfix increment

$Z++ \Rightarrow Z = Z + 1$

```
Z=5;  
X= Z++;
```

x gets z's old value $\rightarrow x = 5$

Then z increases by 1 $\rightarrow z = 6$

2) Prefix increment

$++Z \Rightarrow Z = Z + 1$

```
X =++ Z
```

z increases by 1 $\rightarrow z = 6$

x gets the new value $\rightarrow x = 6$

DECREMENT

1) Postfix Decrement

$Z-- \Rightarrow Z = Z - 1$

```
Z=5;  
X= Z--;
```

x gets z's old value $\rightarrow x = 5$

Then z decrease by 1 $\rightarrow z = 4$

2) Prefix Decrement

$--Z \Rightarrow Z = Z - 1$

```
X =-- Z
```

z decrease by 1 $\rightarrow z = 4$

x gets the new value $\rightarrow x = 4$

Operations Hierarchy Example

$$3 * (6 + 2) / 12 - (7 - 5) ^ 2 * 3$$

() first

$$= 3 * 8 / 12 - 2 ^ 2 * 3$$

^ next

$$= 3 * 8 / 12 - 4 * 3$$

Mult / Div (L to R)

$$= 24 / 12 - 4 * 3$$

Mult / Div (L to R)

$$= 2 - 12$$

Add / Subtr

$$= -10$$

EXAMPLE I

```
#include <stdio.h>

int main() {
    int result1 = 2 + 3 * 4;
    int result2 = (2 + 3) * 4;

    printf("Result1: %d\n", result1);
    printf("Result2: %d\n", result2);

    return 0;
}
```

Output

Result1: 14
Result2: 20

=== Code Execution Successful ===

ASSIGNMENT OPERATORS

- Assignment operators are used to assign values to variables.

Operator	Example	Same As
=	x = 5	x = 5
+=	x += 3	x = x + 3
-=	x -= 3	x = x - 3
*=	x *= 3	x = x * 3
/=	x /= 3	x = x / 3
%=	x %= 3	x = x % 3
^=	x ^= 3	x = x ^ 3

EXAMPLE

```
#include <stdio.h>

int main() {
    int x = 100;
    printf("x: %d\n", x);
    x += 50;
    printf("x: %d\n", x);
    x -= 20;
    printf("x: %d\n", x);
    x *= 2;
    printf("x: %d\n", x);
    x /=2;
    printf("x: %d\n", x);
    x %=2;
    printf("x: %d\n", x);

    return 0;
}
```

Output

```
x: 100
x: 150
x: 130
x: 260
x: 130
x: 0
```

=== Code Execution Successful ===

COMPARISON OPERATORS

- Comparison operators are used to compare two values (or variables).
- The return value of a comparison is either 1 or 0, which means true (1) or false (0). These values are known as Boolean values

Operator	Name	Example
==	Equal to	x == y
!=	Not equal	x != y
>	Greater than	x > y
<	Less than	x < y
>=	Greater than or equal to	x >= y
<=	Less than or equal to	x <= y

EXAMPLE

```
#include <stdio.h>

int main() {
    int age = 18;

    printf("%d\n", age >= 18); // (true), old enough to vote
    printf("%d\n", age < 18);  // (false)
    return 0;
}
```

Output

```
1
0
```

```
=== Code Execution Successful ===
```

LOGICAL OPERATORS

You can also test for true or false values with logical operators. Logical operators are used to determine the logic between variables or values.

Operator	Name	Description	Example
&&	Logical and	Returns true if both statements are true	<code>x < 5 && x < 10</code>
	Logical or	Returns true if one of the statements is true	<code>x < 5 x < 4</code>
!	Logical not	Reverse the result, returns false if the result is true	<code>!(x < 5 && x < 10)</code>

EXAMPLE

```
#include <stdio.h>

int main() {
    int isLoggedIn = 1;
    int isAdmin = 0;

    printf("Regular user: %d\n", isLoggedIn && !isAdmin);
    printf("Has access: %d\n", isLoggedIn || isAdmin);
    printf("Not logged in: %d\n", !isLoggedIn);
    return 0;
}
```

Output

```
Regular user: 1
Has access: 1
Not logged in: 0
```

```
=== Code Execution Successful ===
```

- Calculate the Total Cost of an Item
- **Instructions**
 - Inside main(), complete the following steps: Declare two int variables named itemPrice and shippingCost, and assign them values
 - Create an int variable named sum
 - Calculate the total cost by adding itemPrice and shippingCost (store the result in sum)
 - Print the total cost using printf



Happy Coding!